LECTURE 3

SUMMARY

MODULE 1 –LECTURE 3

MAINTENANCE CALORIES & METABOLISM EXPLAINED IN DETAIL

HOW TO START YOUR FAT LOSS OR MUSCLE GAIN JOURNEY

Beginning a fitness journey isn't just diet + workout. It's a structured, logical, and informed process.
These are the fundamental steps you must understand before entering fat loss or muscle gain.

UNDERSTAND YOUR BODY STRUCTURE

Before doing anything, you must know:

- your body structure
- your symmetry
- your muscle development
- your weak areas
- your conditioning
- your presentation

Somatotype (ectomorph, mesomorph, endomorph) is an outdated theory scientifically—but it's still useful as a visual reference, so you can identify:

- where you store fat
- where you store muscle
- which muscles are developed
- which areas look under-trained

This helps you plan your training better.

IDENTIFY WEAK BODY PARTS

A great transformation comes from fixing weak areas, such as:

- weak shoulders
- weak chest
- lagging back
- underdeveloped legs
- less defined glutes
- poor arms
- weak abs

You need to clearly recognize:

→ “Which muscles do I need to improve first?”

This determines your training priority and exercise selection.

BE VERY CLEAR WITH YOUR AIM

Confusion = no results.

You must decide:

- Are you starting a fat loss phase?
- Or a muscle gain phase?
- Or a recomposition phase?

Your nutrition, training volume, cardio, and recovery all depend on your goal. Be specific.

UNDERSTAND WORKOUT CYCLES (PERIODISATION)

Training is structured in 3 time scales:

• **Microcycle**: 7–14 days

A short block of training
Example: 1–2 week training template.

• **Mesocycle**: 4–8 weeks

A progression block
Example: hypertrophy block, strength block, fat loss block.

• **Macrocycle**: full training period (months to year)

Your entire transformation timeline.

If you keep jumping programs → no long-term adaptation happens.

DON’T CHANGE WORKOUTS TOO FREQUENTLY

Your muscles need consistent stimulus to adapt.

Changing workouts every week kills your progress.

Progress requires:

- repeated training
- repeated overload
- consistency in movements
- tracking progression

If you switch too often, you lose:

- strength progression
- better technique
- measurable improvement

UNDERSTAND TDEE (TOTAL DAILY ENERGY EXPENDITURE)

TDEE = NEAT + TEF + EAT + BMR

Where:

- NEAT: movement during daily life (walking, cleaning)
- TEF: energy used for digesting food
- EAT: energy burnt during training
- BMR: calories your body burns at rest

Your calories for fat loss/gain depend on your TDEE.

BODY FAT LEVELS & HEALTH

MEN

Normal: 15–20%

Athletic: 10–12%

< 10% is ONLY for stage/competition and CANNOT be maintained.

Extremely low → hormone crash, low libido, fatigue, irritability, poor recovery.

WOMEN

Normal: 20–22%

Stage: 12–15%

Very High Body Fat (40%+)

→ Poor insulin sensitivity
→ More calories required just for insulin to function
→ Higher risk of metabolic disease

Both extremes (too low / too high) are unhealthy.

MUSCLE RETENTION DURING FAT LOSS

During fat loss:

- the goal is NOT to lose weight
- the goal is to lose fat while retaining muscle

Muscle retention requires:

- high protein
- resistance training
- proper sleep
- controlled deficit (not aggressive crash diets)
- strength maintenance
- progressive overload when possible

WHAT IS FAT?

Fat is:

- stored energy
- backup fuel system
- insulation
- essential for hormones
- a survival mechanism
- part of cell membranes
- necessary for vitamin absorption (A, D, E, K)

BUT:

- Excess fat → poor insulin sensitivity
- Too little fat → hormonal crash, infertility, poor cognitive function
- Balanced fat levels = optimal performance.

INSULIN SENSITIVITY

Pancreas

- Regulates carbohydrate metabolism through insulin and glucagon.

Glucagon

- Released when blood sugar is too low (hypoglycemia)
- Converts stored glycogen into glucose
- Increases glucose availability
- Reduces liver glucose consumption (to spare glucose for brain)
- In fasting state → uses stored fat for energy

Insulin

- Released when blood sugar rises (hyperglycemia)
- Helps store nutrients
- Moves glucose into muscle/liver
- Controls blood sugar
- Lowers cortisol (reason high-carb meals feel comforting or euphoric)

Insulin Sensitivity

- How easily insulin does its job.

Lean individuals

- → High insulin sensitivity
- → Require fewer calories / less insulin to control blood sugar

Fat individuals

- → Low insulin sensitivity
- → Require more insulin
- → Blood sugar stays high longer
- → Higher risk of metabolic issues

Food Effects

- High protein + low carb meal → increases glucagon
- High carb + low protein meal → increases insulin
- Very high calorie meal → spikes insulin strongly